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The Molecular Structure of Thought: Mapping the Topology of Long Chain-of-Thought Reasoning

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Abstract

Large language models (LLMs) often fail to learn effective long chain-of-thought (Long CoT) reasoning from human or non-Long-CoT LLMs imitation. To understand this, we propose that effective and learnable Long CoT trajectories feature stable molecular-like structures in unified view, which are formed by three interaction types: Deep-Reasoning (covalent-like), Self-Reflection (hydrogen-bond-like), and Self-Exploration (van der Waals-like). Analysis of distilled trajectories reveals these structures emerge from Long CoT fine-tuning, not keyword imitation. We introduce Effective Semantic Isomers and show that only bonds promoting fast entropy convergence support stable Long CoT learning, while structural competition impairs training. Drawing on these findings, we present Mole-Syn, a distribution-transfer-graph method that guides synthesis of effective Long CoT structures, boosting performance and RL stability across benchmarks.

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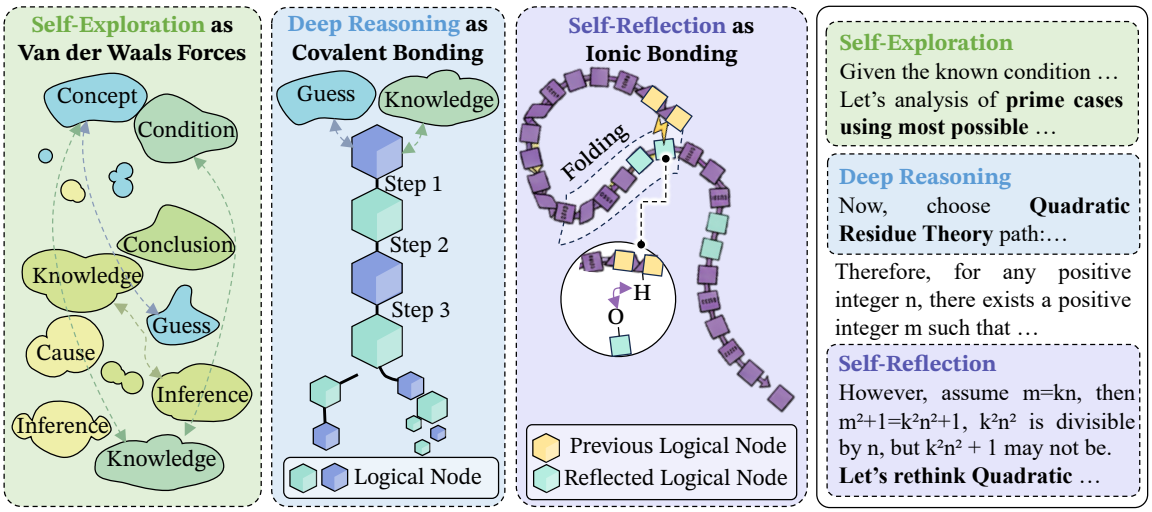

Figure 1 The hypothesis that stable molecular structure in Long CoT arises from three key "chemical" bonds.

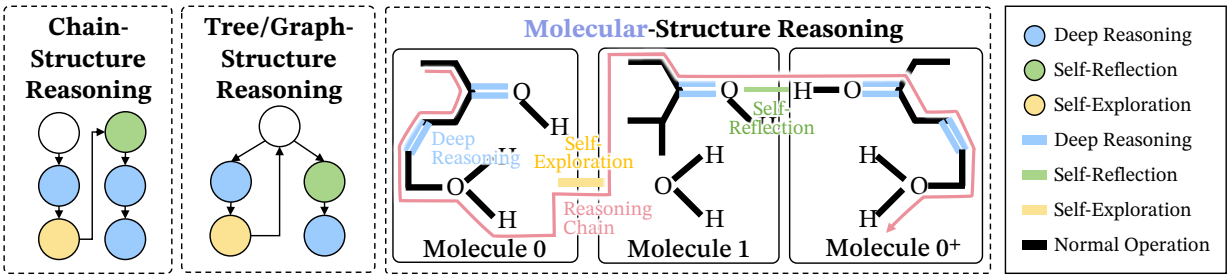

Figure 2 Comparison of prior chain- or tree-like structures and our molecular structure. Reasoning starts from Mole. 0, uses **deep reasoning** on strongly related structures, then employs **self-exploration** for new logic in Mole. 1. When meet errors, reasoning utilize **self-reflection** to guide chain to optimized Mole. 0^+.

1 Introduction

Recently, large language models (LLMs) have excelled on diverse reasoning tasks via explicit chain-of-thought (CoT) rationales [1–4]. Yet, they struggle to cold-start from instruction-tuned or base models into Long CoT models requiring extended multi-step reasoning [5, 6]. Notably, Du et al. [7] shows that humans generate Long CoT rationales without imitating DeepSeek-R1 [8]. Our preliminary studies reveal that standard supervised fine-tuning and distillation from human or Instruction LLM rationales (using randomly sampled Long CoT examples) fail to reliably instill these skills in LLMs. Models often lose coherence over long trajectories or fail to transfer patterns to novel tasks. This prompts a key question:

How do Large Language Models learn and represent effective Long Chain-of-Thought?

To explain this, we posit that they acquire the organization of reasoning trajectories. As shown in Figure 2, prior studies model these as logic nodes in sequences or trees of steps. Yet, our analysis of Long CoT across strong reasoning models reveals a stable distribution of three core behaviors across tasks and architectures: Deep-Reasoning, Self-Reflection, and Self-Exploration [5], which node-centric views fail to capture.

This finding triggers a molecular-inspired, distributional view: we model behavior-labeled logic edges as interaction bonds and examine how their global **molecular-like structure** ensures long-horizon reasoning stability. Specifically, Deep-Reasoning forms dense local clusters of coupled deductions, like covalent bonds; Self-Reflection creates long-range corrective links to prior steps, like hydrogen bonds; and Self-Exploration forges weak bridges between distant clusters, like van der Waals forces. Thus, high-quality Long CoT arises from the **stable composition and arrangement of these bond types**, guiding effective learning*.

In this framework, we define semantic isomers as Long CoT trajectories that solve the same tasks and visit similar semantic regions but differ in behavior distributions and transitions. We demonstrate that multiple near-optimal semantic isomers exist per task family, but mixing stable isomers from different strong teachers destabilizes learning, degrading performance and behavior distributions despite matched token statistics. This structurally explains why combining heterogeneous Long CoT traces often fails, beyond token-level distillation.

Building on this perspective, we propose Mole-Syn, a structure-aware synthesis framework that first estimates a behavior transition graph from strong reasoning models and then transfers only this behavioral structure to cheaper instruction LLMs via controlled trajectory synthesis, instead of directly copying teacher outputs. This decouples structural transfer from model-specific surface form, enables the generation of Long CoT data that match target behavior distributions from scratch, and yields consistent gains in both Long CoT performance and RL stability across six benchmarks.

After that, we analyze the shaping function of each bond in the Long CoT structure. Deep Reasoning bonds encode core logical flow, Self-Reflection bonds support folding pathways to previous steps, and Self-Exploration bonds reinforce long-range consistency checks, enabling targeted bond distributions. Moreover, we discuss why a deteriorated molecular structure is hard to restore, which helps explain how private LLMs protect Long

*Note: C, H, and O atom references are analogies for molecular structure only.

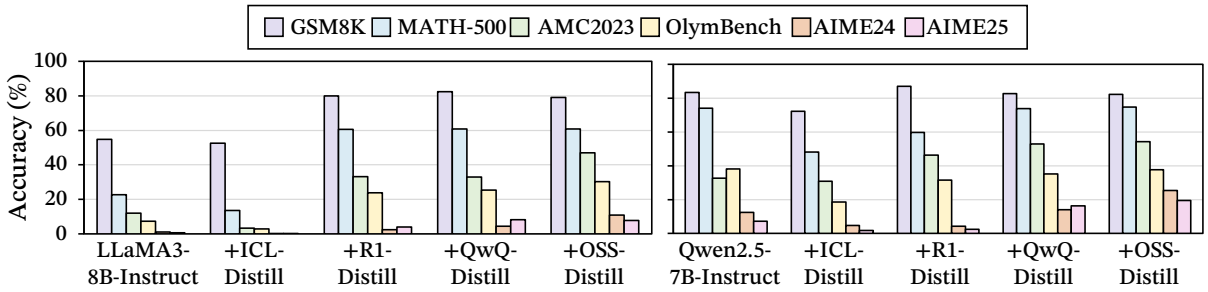

Figure 3 The failure of distillation from weak instruction LLMs with ICL and Human-annotated reasoning traces to acquire Long CoT structures, compared to successful distillation from strong reasoning LLMs. See Appendix Figure 15 for the full result.

CoT structures from distillation-based imitation. Methods such as summarization and reasoning compression can disrupt Long CoT structure, limiting unauthorized replication of internal reasoning processes.

In summary, our contributions are as follows:

- We model Long CoT as a molecular structure with 3 bonds: deep-reasoning (covalent), self-reflection (hydrogen-bond), and self-exploration (van der Waals), to understand its effective learning.
- We identify effective Semantic Isomers for Long CoT learning, where only entropy-convergent bonds enable stable learning, while competing structures destabilize learning.
- We introduce Mole-Syn, which uses distribution-transfer graphs to synthesize these structures, improving Long CoT and stabilizing RL across 6 benchmarks.

2 Preliminary: Cold-start LLMs for Long Chain-of-Thought

First, we validate whether LLMs exhibit valid Long CoT trajectories suitable for Long CoT learning. We investigate three data sources: distillation from strong reasoning LLMs, distillation from weak instruction LLMs with in-context learning (ICL), and fine-tuning on human reasoning traces.

Only distillation from strong reasoning LLMs works. To identify effective data sources, we curated a synthetic set of reasoning traces from three sources. As shown in Figure 3, only distillation from strong reasoning LLMs enables target models to learn and retain Long CoT structure, improving performance on benchmarks requiring extended reasoning. These results indicate that only high-quality traces reliably support both learning and use of Long CoT structures.

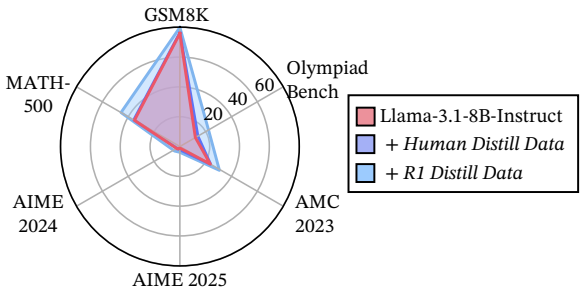

Figure 4 Performance comparison between human-annotated reasoning traces (*+ Human Distill Data*) and R1 distilled reasoning traces (*+ R1 Distill Data*).

Distillation from randomly selected ICL by weak instruct LLMs does not work. Instruction LLMs without explicit reasoning training lag behind reasoning-specialized models. We test whether LLMs can acquire Long CoT structure by distilling from an instruction LLM using randomly selected ICL demonstrations that emulate Long CoT reasoning. As shown in Figure 3, performance drops sharply. Instruction LLMs can only mimic short CoT traces (∼6–8 steps) and fail to extend exploration while preserving intermediate steps and trace coherence. This degradation indicates a limitation of ICL-based distillation rather than robust long-chain imitation.